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$$f(x) = 2x^2 - x \quad a \in \{-3, 0, 2\}$$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$f(a+h) = 2(a+h)^2 - (a+h)$$

$$= 2(a^2 + 2ah + h^2) - a - h$$

$$= 2a^2 + 4ah + 2h^2 - a - h$$

$$f(a+h) - f(a) = 2a^2 + 4ah + 2h^2 - a - h - (2a^2 - a)$$

$$= 4ah + 2h^2 - h$$

$$\frac{f(a+h) - f(a)}{h} = \frac{4ah + 2h^2 - h}{h}$$

$$= 4a + 2h - 1$$

$$f'(a) = \lim_{h \to 0} (4a + 2h - 1)$$

$$= 4a - 1$$

$$f'(-3) = 4(-3) - 1 = -13$$

$$f'(0) = 4(0) - 1 = -1$$

$$f'(2) = 4(2) - 1 = 7$$

References